

## Midterm 2 Review class - November 14

Problem 1. You are selling cars and giving the following warranty. When the buyer asks you to repair the car for the first time, he will give you 1000\$. If he comes for the second time you will return him these 1000\$ and give him 1000\$ more. For every subsequent repair no money is exchanged like this. The 1000\$ amounts all go from and into the “secret account”, and your goal is to maximize the expected gain on this secret account for every car you sell. You know that the number of times buyer comes to repair the car is a Poisson random variable with parameter  $\lambda$ . What is the expected gain in the secret account for every car you sell? Which  $\lambda$  would maximize this value.

**Solution:** Let  $X$  be number of repairs of a car and  $Y$  the income into the “secret account”. If there are no repairs there will be no money exchange so  $Y = 0$ , if  $X = 1$  then we get 1000\$ from the customer, so  $Y = 1000$ , if  $X = 2$  then we will first get 1000\$ (during the 1st repair), and then return this 1000\$ and pay 1000\$ more, so in this case  $Y = -1000$ . This is also the case if  $X \geq 3$ . Therefore we have  $\mathbb{E}(Y|X = 0) = 0$ ,  $\mathbb{E}(Y|X = 1) = 1000$  and  $\mathbb{E}(Y|X \geq 2) = -1000$  and

$$\begin{aligned}\mathbb{E}(Y) &= \mathbb{P}(X = 0)\mathbb{E}(Y|X = 0) + \mathbb{P}(X = 1)\mathbb{E}(Y|X = 1) + \mathbb{P}(X \geq 2)\mathbb{E}(Y|X \geq 2) \\ &= 1000\mathbb{P}(X = 1) - 1000\mathbb{P}(X \geq 2) = 1000\lambda e^{-\lambda} - 1000(1 - e^{-\lambda} - \lambda e^{-\lambda}) \\ &= 1000(2\lambda e^{-\lambda} + e^{-\lambda} - 1).\end{aligned}$$

To maximize this value we differentiate the expression on the right hand side with respect to  $\lambda$  to get

$$1000(2e^{-\lambda} - 2\lambda e^{-\lambda} - e^{-\lambda}) = 1000e^{-\lambda}(1 - 2\lambda).$$

The derivative is positive for  $\lambda < 1/2$  equal to zero at  $\lambda = 1/2$  and negative for  $\lambda > 1/2$ . Therefore, the maximum is attained at  $\lambda = 1/2$ .

Problem 2. Pair of random variables  $(X, Y)$  has range

$$\{(2^{-k}, 2^{-l}) | l, k \text{ are integers and } 1 \leq k < l\}.$$

The joint probability mass function is given by

$$p_{X,Y}(2^{-k}, 2^{-l}) = C2^{-k-l},$$

for some constant  $C$ . Find the value of  $C$ . Find the marginal probability mass functions for both  $X$  and  $Y$  and evaluate  $\mathbb{E}(X)$  and  $\mathbb{E}(Y)$ .

**Solution:** We get  $C$  from the condition that the probabilities have to add up to 1. This gives

$$\sum_{k=1}^{\infty} \left( \sum_{l=k+1}^{\infty} C2^{-k-l} \right) = 1.$$

The inner sum  $\sum_{l=k+1}^{\infty} C2^{-k-l}$  is the sum of geometric series whose first element is  $C2^{-2k-1}$  and the step factor is  $1/2$  so

$$\sum_{l=k+1}^{\infty} C2^{-k-l} = \frac{C2^{-2k-1}}{1 - 1/2} = C4^{-k}$$

Now the condition is  $\sum_{k=1}^{\infty} \frac{C}{4^k} = 1$  that is

$$\frac{C/4}{1 - 1/4} = 1 \Rightarrow C = 3.$$

The range of  $X$  is  $2^{-k}$  for integers  $k \geq 1$  and the marginal PMF of  $X$  is

$$\mathbb{P}(X = 2^{-k}) = \sum_{l=k+1}^{\infty} \mathbb{P}(X = 2^{-k}, Y = 2^{-l}) = 3 \sum_{l=k+1}^{\infty} 2^{-k-l}.$$

This is geometric series whose first element is  $2^{-2k-1}$  and step factor  $1/2$  so

$$\mathbb{P}(X = 2^{-k}) = 3 \frac{2^{-2k-1}}{1 - 1/2} = \frac{3}{4^k}.$$

The range of  $Y$  is  $2^{-l}$  for  $l \geq 2$  and the marginal PMF of  $Y$  is

$$\mathbb{P}(Y = 2^{-l}) = \sum_{k=1}^{l-1} \mathbb{P}(X = 2^{-k}, Y = 2^{-l}) = 3 \sum_{k=1}^{l-1} 2^{-k-l} = 3 \cdot 2^{-l-1} \left(1 + \frac{1}{2} + \dots + \frac{1}{2^{l-2}}\right)$$

The expression in the sum is equal to  $\frac{1 - 1/2^{l-1}}{1 - 1/2} = 2 - 2^{-l+2}$ , so  $\mathbb{P}(Y = 2^{-l}) = 3 \cdot 2^{-l} - 3 \cdot 2^{-2l+1}$ .

Now the moments are

$$\mathbb{E}(X) = \sum_{k=1}^{\infty} 2^{-k} \mathbb{P}(X = 2^{-k}) = \sum_{k=1}^{\infty} \frac{3}{8^k} = \frac{3/8}{1 - 1/8} = \frac{3}{7},$$

and

$$\mathbb{E}(Y) = \sum_{l=2}^{\infty} 2^{-l} \mathbb{P}(Y = 2^{-l}) = \sum_{l=2}^{\infty} 3 \cdot 4^{-l} - \sum_{l=2}^{\infty} 6 \cdot 8^{-l} = \frac{3/16}{1 - 1/4} - \frac{6/64}{1 - 1/8} = \frac{1}{4} - \frac{3}{28} = \frac{1}{7}.$$

**Problem 3.** We have  $n$  guests and  $n$  different locked boxes. Assume that  $k$ -th guest has the keys to open the first  $k$  boxes. In other words, the first guest has only the key for the first box, the second guest has the keys to first two boxes, ..., the last guest has all the keys. The  $n$  boxes are distributed among the  $n$  guests uniformly at random (each guest gets exactly one box). What is the probability that the  $k$ -th guest opens his box? What is the expected number of boxes that can be opened by guests?

**Solution:** The  $k$ -th guest will open the box (s)he gets if it is one of the first  $k$  boxes. The probability of this is  $k/n$ . Let  $X_k$  be the indicator variable of this event (that is it has value 1 if  $k$ -th person opens the box and 0 otherwise). Then  $\mathbb{E}(X_k) = \mathbb{P}(X_k = 1) = k/n$ . Then the total number of open boxes is

$$X = X_1 + X_2 + \dots + X_n.$$

The expectation is

$$\mathbb{E}(X) = \mathbb{E}(X_1) + \mathbb{E}(X_2) + \dots + \mathbb{E}(X_n) = \frac{1}{n} (1 + 2 + \dots + n) = \frac{n(n+1)}{2n} = \frac{n+1}{2}.$$

**Problem 4.** Let  $X$  be a standard Normal random variable, and  $Y$  a geometric random variable with parameter  $1/4$ . If  $X$  and  $Y$  are independent compute  $\mathbb{E}(e^X)$ ,  $\mathbb{E}(e^Y)$  and  $\mathbb{E}(e^{X+Y})$ .